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Manufacturing method of semiconductor device and semiconductor wafer

Abstract

A manufacturing method of a semiconductor device includes preparing a silicon carbide substrate, growing an epitaxial layer, and forming a structure. The silicon carbide substrate has an upper surface on which an alignment mark having a recessed shape is disposed, and a perpendicular line to the upper surface is inclined with respect to a [0001] direction toward a [11-20] direction. The epitaxial layer is grown on the upper surface and covers the alignment mark. The structure is formed on or above the upper surface at a position apart from the alignment mark by an interval P in the [11-20] direction along the upper surface. The interval P satisfies a relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is a depth of the alignment mark and θ is an inclination angle of the perpendicular line with respect to the [0001] direction.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims the benefit of priority from Japanese Patent Application No. 2022-035666 filed on Mar. 8, 2022. The entire disclosure of the above application is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a manufacturing method of a semiconductor device and also relates to a semiconductor wafer.

BACKGROUND

(3) Conventionally, there has been known a silicon carbide (SiC) substrate that has an upper surface on which an alignment mark having a recessed shape is disposed.

SUMMARY

(4) According to an aspect of the present disclosure, a manufacturing method of a semiconductor device includes preparing a SiC substrate, growing an epitaxial layer, and forming a structure. The SiC substrate has an upper surface on which an alignment mark having a recessed shape is disposed, and a perpendicular line that is perpendicular to the upper surface of the SiC substrate is inclined with respect to a [0001] direction toward a [11-20] direction. The epitaxial layer is grown on the upper surface of the SiC substrate so as to cover the alignment mark. The structure is formed on or above the upper surface of the SiC substrate at a position apart from the alignment mark by an interval P in the [11-20] direction along the upper surface of the SiC substrate. The interval P satisfies a relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is a depth of the alignment mark and θ is an inclination angle of the perpendicular line with respect to the [0001] direction.

(5) According to another aspect of the present disclosure, a semiconductor wafer includes a base substrate, an epitaxial layer, and a structure. The base substrate is made of silicon carbide and has an upper surface on which an alignment mark having a recessed shape is disposed. A perpendicular line that is perpendicular to the upper surface of the base substrate is inclined with respect to a [0001] direction toward a [11-20] direction. The epitaxial layer is disposed on the upper surface of the base substrate and covers the alignment mark. The structure is disposed on the upper surface of the base substrate or an upper surface of the epitaxial layer. The structure is disposed at a position apart from the alignment mark by an interval P in the [11-20] direction along the upper surface of the base substrate. The interval P satisfies a relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is a depth of the alignment mark and θ is an inclination angle of the perpendicular line with respect to the [0001] direction.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Objects, features and advantages of the present disclosure will become apparent from the following detailed description made with reference to the accompanying drawings. In the drawings:

(2) FIG. 1 is a plan view of a SiC substrate;

(3) FIG. 2 is a cross-sectional view of the SiC substrate;

(4) FIG. 3 is a plan view of alignment marks on an upper surface of the SiC substrate;

(5) FIG. 4 is a cross-sectional view illustrating a first manufacturing method of the alignment marks;

(6) FIG. 5 is a cross-sectional view illustrating a process in a second forming method of the alignment marks;

(7) FIG. 6 is a cross-sectional view illustrating a process in the second forming method of the alignment marks, subsequent to the process illustrated in FIG. 5;

(8) FIG. 7 is a cross-sectional view illustrating a process in the second forming method of the alignment marks, subsequent to the process illustrated in FIG. 6;

- (9) FIG. 8 is a cross-sectional view illustrating a process in a third forming method of the alignment marks;
- (10) FIG. 9 is a cross-sectional view illustrating a process in the third forming method of the alignment marks, subsequent to the process illustrated in FIG. 8;
- (11) FIG. 10 is a cross-sectional view illustrating a process in the third forming method of the alignment marks, subsequent to the process illustrated in FIG. 9;
- (12) FIG. 11 is an enlarged cross-sectional view of the alignment marks;
- (13) FIG. 12 is a plan view of alignment marks on an upper surface of a SiC layer;
- (14) FIG. 13 is a cross-sectional view illustrating a process in an epitaxial growth process;
- (15) FIG. 14 is a cross-sectional view illustrating a process in the epitaxial growth process, subsequent to the process illustrated in FIG. 13;
- (16) FIG. 15 is a cross-sectional view illustrating a process in the epitaxial growth process, subsequent to the process illustrated in FIG. 14;
- (17) FIG. 16 is a cross-sectional view illustrating a process in the epitaxial growth process, subsequent to the process illustrated in FIG. 15;
- (18) FIG. 17 is an enlarged cross-sectional view illustrating an alignment mark according to a second embodiment;
- (19) FIG. 18 is a cross-sectional view illustrating an epitaxial growth process according to the second embodiment;
- (20) FIG. 19 is a cross-sectional view illustrating a structure formation process according to the second embodiment;
- (21) FIG. 20 is a cross-sectional view illustrating a structure formation process according to a modification of the second embodiment; and
- (22) FIG. 21 is a plan view illustrating an alignment mark formed on an upper surface of a SiC substrate according to a modification.

DETAILED DESCRIPTION

(23) A SiC substrate may have an upper surface on which an alignment mark having a recessed shape is disposed. In the SiC substrate, the upper surface may be inclined with respect to a (0001) plane so that an off direction is a [11-20] direction. In other words, in the SiC substrate, a perpendicular line that is perpendicular to the upper surface of the SiC substrate may be inclined with respect to a [0001] direction toward the [11-20] direction. When an epitaxial layer is grown on the upper surface of the SiC substrate so as to cover the alignment mark, a facet surface inclined with respect to the upper surface of the SiC substrate may be formed at a position adjacent to the alignment mark in the [11-20] direction. The formation of the facet surface may be restricted by forming the alignment mark into a predetermined shape.

(24) However, there may be cases where formation of a facet surface is not prevented even by forming the alignment mark into the predetermined shape. Moreover, due to various restrictions, it may not be possible to form an alignment mark into the predetermined shape and prevent formation of a facet surface. After an epitaxial layer is formed so as to cover the alignment mark, alignment may be performed using a recessed portion on an upper surface of the epitaxial layer (that is, a recessed portion formed following the alignment mark). At this time, if a facet surface formed on the epitaxial layer interferes with other structures, an alignment apparatus cannot correctly recognize the recessed portion on the upper surface of the epitaxial layer, and proper alignment cannot be performed.

(25) According to an aspect of the present disclosure, a manufacturing method of a semiconductor device includes preparing a SiC substrate, growing an epitaxial layer, and forming a structure. The SiC substrate has an upper surface on which an alignment mark having a recessed shape is disposed, and a perpendicular line that is perpendicular to the upper surface of the SiC substrate is inclined with respect to a [0001] direction toward a [11-20] direction. The epitaxial layer is grown on the upper surface of the SiC substrate so as to cover the alignment mark. The structure is formed

on or above the upper surface of the SiC substrate at a position apart from the alignment mark by an interval P in the [11-20] direction along the upper surface of the SiC substrate. The interval P satisfies a relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is a depth of the alignment mark and θ is an inclination angle of the perpendicular line with respect to the [0001] direction.

(26) Note that the preparing the SiC substrate may include bringing the SiC substrate with the alignment mark in a manufacturing facility or forming the alignment mark on the upper surface of the SiC substrate.

(27) The forming the structure on or above the upper surface of the SiC substrate may be performed before or after the growing the epitaxial layer. In other words, the structure may be formed on the upper surface of the SiC substrate before the forming the epitaxial layer or the structure may be formed above the upper surface of the SiC substrate after the forming the epitaxial layer (that is, on an upper surface of the epitaxial layer).

(28) According to experiments by the inventors, it has been found that a width of a facet surface formed adjacent to the alignment mark when the epitaxial layer is grown to cover the alignment mark can be estimated by the inclination angle θ (so-called, an off angle) of the upper surface of the SiC substrate with respect to the direction and the depth D of the alignment mark. In general, the width of the facet surface is narrower than a width W calculated by an equation of $W = D/\tan \theta$. Therefore, by setting the interval P so as to satisfy a relationship of $D/\tan \theta < P$, it is possible to restrict the facet surface from interfering with the structure. If the interval P is unnecessarily widened, an area occupied by the interval P becomes wider, which causes a decrease in the number of semiconductor devices manufactured from the SiC substrate. As described above, by setting the interval P so as to satisfy the relationship of $D/\tan \theta < P < 10D/\tan \theta$ (that is, $W < P < 10W$), interference between the facet surface and the structure can be restricted without unnecessarily widening the interval P. Since the interference between the facet surface and the structure can be restricted, it is possible to perform proper alignment using the recessed portion on the upper surface of the epitaxial layer (that is, the recessed portion formed following the alignment mark).

(29) According to another aspect of the present disclosure, a semiconductor wafer includes a base substrate, an epitaxial layer, and a structure. The base substrate is made of silicon carbide and has an upper surface on which an alignment mark having a recessed shape is disposed. A perpendicular line that is perpendicular to the upper surface of the base substrate is inclined with respect to a [0001] direction toward a [11-20] direction. The epitaxial layer is disposed on the upper surface of the base substrate and covers the alignment mark. The structure is disposed on the upper surface of the base substrate or an upper surface of the epitaxial layer. The structure is disposed at a position apart from the alignment mark by an interval P in the [11-20] direction along the upper surface of the base substrate. The interval P satisfies a relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is a depth of the alignment mark and θ is an inclination angle of the perpendicular line with respect to the [0001] direction.

(30) According to the semiconductor wafer described above, it is possible to perform proper alignment using the recessed portion on the upper surface of the epitaxial layer.

First Embodiment

(31) In a manufacturing method of a first embodiment, a semiconductor device is manufactured from a SiC substrate **12** illustrated in FIG. 1 and FIG. 2. The SiC substrate **12** is made of 4H—SiC. Hereinafter, a thickness direction of the SiC substrate **12** is referred to as a z-direction, one direction perpendicular to the z-direction is referred to as an x-direction, and a direction perpendicular to the z-direction and the x-direction is referred to as a y-direction. The x-direction and the y-direction are parallel to an upper surface **12a** of the SiC substrate **12**. As illustrated in FIG. 2, a [1-100] direction of the SiC substrate **12** matches the y-direction. A (0001) plane of the SiC substrate **12** is inclined by an off angle θ around the [1-100] direction with respect to the upper surface **12a** of the SiC substrate **12**. In other words, the (0001) plane is inclined with respect to the upper surface **12a** in a cross section including a [11-20] direction and a direction. A perpendicular

line **12s** in FIG. 2 indicates a perpendicular line that is perpendicular to the upper surface **12a**. Since the (0001) plane is inclined by the off angle θ with respect to the upper surface **12a**, the perpendicular line **12s** is inclined with respect to the [0001] direction by the off angle θ toward the [11-20] direction. Note that the off angle θ may be 10 degrees or less, or may be 5 degrees or less. For example, the off angle θ can be approximately 4 degrees. As illustrated in FIG. 1, when the upper surface **12a** of the SiC substrate **12** is viewed from above, the x direction extends along the [11-20] direction.

(32) In the manufacturing method of the first embodiment, first, an alignment mark formation process is performed. In the alignment mark formation process, alignment marks **20** illustrated in FIG. 3 are formed on the upper surface of the SiC substrate **12**. Each of the alignment marks **20** has a recessed shape disposed on the upper surface **12a** of the SiC substrate **12**. Each of the alignment marks **20** has long sides extending in the y-direction and short sides extending in the x-direction. Therefore, when the upper surface **12a** of the SiC substrate **12** is viewed from above, a side surface **24** (that is, an edge portion) of each of the alignment marks **20** located in the [11-20] direction is perpendicular to the [11-20] direction. The alignment marks **20** are arranged with an interval P therebetween in the x-direction. Although not illustrated in FIG. 1, the upper surface **12a** of the SiC substrate **12** has an element region in which a semiconductor element structure is formed and an outer region outside the element region. The outer region is a region that is removed when the SiC substrate **12** is divided into multiple semiconductor devices by dicing or the like. Each of the alignment marks **20** is formed in the outer region of the upper surface **12a** of the SiC substrate **12**.

(33) In the alignment mark formation process, the alignment marks **20** can be formed by various methods. A first forming method of the alignment marks **20** will be described with reference to FIG. 4. In the first forming method, first, as illustrated in FIG. 4, a resist mask **30** having opening portions **30a** is formed on the upper surface **12a** of the SiC substrate **12**. Next, the upper surface **12a** of the SiC substrate **12** is etched by anisotropic etching through the opening portions **30a**. Accordingly, the alignment marks **20** each having the recessed shape are formed in the respective opening portions **30a**. After forming the alignment marks **20**, the resist mask **30** is removed.

(34) A second forming method of the alignment marks **20** will be described with reference to FIGS. 5 to 7. FIGS. 5 to 7 illustrate cross sections of the element region **13** and the outer region **14**. In the second forming method, first, as illustrated in FIG. 5, a hard mask **32** (for example, a mask made of silicon oxide) is formed on the upper surface **12a** of the SiC substrate **12**. The hard mask **32** is a mask for controlling a range of ion implantation into SiC substrate **12**. Next, a resist mask **34** having opening portions **34a** is formed on the hard mask **32**. Then, as illustrated in FIG. 6, the hard mask **32** is etched by anisotropic etching through the opening portions **34a**. Accordingly, opening portions **32a** are formed in the hard mask **32**. Since the opening portions **32a** are formed to reach the SiC substrate **12**, the upper surface **12a** of the SiC substrate **12** is over-etched when forming the opening portions **32a**. Accordingly, recessed shapes are formed on the upper surface **12a** of the SiC substrate **12**. The recessed shapes formed in the outer region **14** become the alignment marks **20**. After forming the opening portions **32a** in the hard mask **32**, the resist mask **34** is removed. Next, n-type or p-type impurity ions are implanted into the upper surface **12a** of the SiC substrate **12** through the hard mask **32**. Accordingly, as illustrated in FIG. 7, a diffusion layer **40** is formed in the SiC substrate **12** below the opening portions **32a**. In the element region **13**, the diffusion layer **40** is formed at required positions. In the outer region **14**, the diffusion layer **40** is formed below the alignment marks **20**. Even if the diffusion layer **40** is formed under the alignment marks **20**, there is no particular problem. As described above, in the second forming method, the alignment marks **20** are formed by utilizing the phenomenon that the upper surface **12a** of the SiC substrate **12** is over-etched when the opening portions **32a** are formed in the hard mask **32** for ion implantation.

(35) A third forming method of the alignment marks **20** will be described with reference to FIGS. 8 to 10. FIGS. 8 to 10 illustrate cross sections of the element region **13** and the outer region **14**. In the third forming method, first, as illustrated in FIG. 8, a resist mask **36** having opening portions **36a** is

formed on the upper surface **12a** of the SiC substrate **12**. Next, n-type or p-type impurity ions are implanted into the upper surface **12a** of the SiC substrate **12** through the resist mask **36**.

Accordingly, as illustrated in FIG. **9**, a diffusion layer **40** is formed in the SiC substrate **12** below the opening portions **36a**. In the element region **13**, the diffusion layer **40** is formed at required positions. Also, in the outer region **14**, the diffusion layer **40** is formed at a position where the alignment marks **20** are to be formed. Next, the SiC substrate **12** is etched by anisotropic etching through the opening portions **36a**. Accordingly, as illustrated in FIG. **10**, recessed shapes are provided on the upper surface **12a** of the SiC substrate **12**. The recessed shapes provided in the outer region **14** become the alignment marks **20**. After forming the alignment marks **20**, the resist mask **36** is removed.

(36) FIG. **11** illustrates a cross-sectional view along line XI-XI of FIG. **3**. In other words, FIG. **11** illustrates a cross-sectional view of alignment marks **20a** and **20b** of the alignment marks **20** along the x-direction. Although the alignment marks **20a** and **20b** are illustrated in FIG. **11**, the other alignment marks **20** are also formed in the same shape as illustrated in FIG. **11**. As illustrated in FIG. **11**, a depth D of each of the alignment marks **20** is smaller than a width Wa of each of the alignment marks **20** in the x-direction. An interval P is provided between each of the alignment marks **20** in the x-direction.

(37) In the manufacturing method of the first embodiment, an epitaxial growth process is performed after the alignment mark formation process. In the epitaxial growth process, a SiC layer **50** illustrated in FIG. **13** and the like is epitaxially grown on the upper surface **12a** of the SiC substrate **12**. The SiC layer **50** is formed to cover each of the alignment marks **20**. When each of the alignment marks **20** is covered with the SiC layer **50**, recessed shapes following the recessed shapes of the alignment marks **20** are formed on the upper surface of the SiC layer **50**. The recessed shapes following the shapes of the alignment marks **20** and formed on the upper surface of the SiC layer **50** are hereinafter referred to as alignment marks **60**. As illustrated in FIG. **12**, a facet surface **60F** is formed at a part of each of the alignment marks **60**. Formation of the facet surface **60F** will be described below.

(38) FIGS. **13** to **16** illustrate how the SiC layer **50** is epitaxially grown on the upper surface **12a** of the SiC substrate **12** in the epitaxial growth process. In the following description, the SiC substrate **12** before the epitaxial growth process is referred to as a base substrate **12b**, and the entirety of the base substrate **12b** and the SiC layer **50** is referred to as the SiC substrate **12**. As illustrated in FIG. **13**, when the SiC layer **50** is grown on the base substrate **12b**, the alignment marks **60** having recessed shapes following the recessed shapes of the alignment marks **20** are formed on the upper surface of the SiC layer **50**. In the following description, the alignment mark **60** following the recessed shape of the alignment mark **20a** may be referred to as an alignment mark **60a**, and the alignment mark **60** following the recessed shape of the alignment mark **20b** may be referred to as an alignment mark **60b**. As illustrated in FIG. **13**, the SiC layer **50** is grown to have a substantially uniform film thickness in a range extending over the upper surface **12a** of the base substrate **12b**, side surfaces **22** of the alignment marks **20** located in the $[-1-120]$ direction, and bottom surfaces **23** of the alignment marks **20**. Hereinafter, a film thickness of the SiC layer **50** on the upper surface **12a** of the base substrate **12b** (more specifically, a film thickness of the portion where the film thickness is uniform) is referred to as film thickness T. The SiC layer **50** hardly grows on the side surfaces **24** of the alignment marks **20** located in the $[11-20]$ direction. Therefore, the SiC layer **50** does not grow uniformly on the upper surface **12a** within a range adjacent to the alignment marks **20** in the $[11-20]$ direction. In this range, the SiC layer **50** grows such that the film thickness of the SiC layer **50** increases with increase in the distance from the alignment marks **20**, and a surface of the SiC layer **50** becomes parallel to the (0001) plane. The surface of the SiC layer **50** parallel to the (0001) plane is the facet surface **60F**.

(39) When the film thickness T of the SiC layer **50** is further increased from a state illustrated in FIG. **13**, a state illustrated in FIG. **14** is obtained. In FIG. **14**, a virtual line **50x** represents the

surface of the SiC layer **50** in the state of FIG. **13**, and a film thickness increase amount $\Delta T1$ indicates a film thickness increased from the state of FIG. **13**. As illustrated in FIG. **14**, even if the film thickness T increases, the SiC layer **50** does not grow on the original facet surface **60F**. Further, when the film thickness T increases, the SiC layer **50** grows so that the surface of the SiC layer **50** is parallel to the (0001) plane at a position adjacent to the original facet surface **60F** in the [11-20] direction. That is, the facet surface **60F** expands in the [11-20] direction. A width Wf in FIGS. **13** and **14** indicates the width of the facet surface **60F** in the x-direction. As illustrated in FIGS. **13** and **14**, the width Wf of the facet surface **60F** increases with increase in the film thickness T . Thus, when the film thickness T is smaller than the depth D of the alignment marks **20**, the width Wf of the facet surface **60F** increases with increase in the film thickness T . In this case, the width Wf of the facet surface **60F** relatively accurately satisfies the relationship of $Wf=T/\tan \theta$.

(40) When the film thickness T is further increased from the state of FIG. **14**, the film thickness T reaches the depth D of the alignment marks **20** as illustrated in FIG. **15**. Then, the alignment marks **20** are filled with the SiC layer **50**. In this state (that is, $T=D$), the width Wf of the facet surface **60F** relatively accurately satisfies the relationship of $Wf=D/\tan \theta$.

(41) When the film thickness T is further increased from the state illustrated in FIG. **15**, the SiC layer **50** grows as illustrated in FIG. **16**. In FIG. **16**, a virtual line **50y** represents the surface of the SiC layer **50** in the state of FIG. **15**, and a thickness film increase amount $\Delta T2$ indicates a film thickness increased from the state of FIG. **15**. As indicated by the film thickness increase amount $\Delta T2$ in FIG. **16**, the SiC layer **50** grows substantially uniformly over the entire surface after the alignment marks **20** are filled. In other words, the SiC layer **50** grows on the facet surface **60F** with a uniform film thickness similar to that on the other surfaces. When the SiC layer **50** grows in this way, even if the film thickness T of the SiC layer **50** increases, the width Wf of the facet surface **60F** hardly changes. Therefore, when the film thickness T is larger than the depth D , even if the film thickness T increases, the width Wf of the facet surface **60F** hardly changes. Therefore, when the film thickness T is larger than the depth D , the width Wf of the facet surface **60F** relatively accurately satisfies the relationship $Wf=D/\tan \theta$.

(42) As described above, when the film thickness T is smaller than the depth D (that is, in the states of FIGS. **13** and **14**), the width Wf of the facet surface **60F** can be relatively accurately satisfies the relationship of $Wf=T/\tan \theta$. In this case, since $T<D$, the width Wf relatively accurately satisfies the relationship $Wf<D/\tan \theta$. Also, when the film thickness T is larger than or equal to the depth D (that is, in the states of FIGS. **15** and **16**), the width Wf relatively accurately satisfies the relationship $Wf=D/\tan \theta$. Therefore, normally, regardless of the film thickness T , the width Wf is not larger than the width W calculated by the equation of $W=D/\tan \theta$. Therefore, by making the interval P in the x-direction between the alignment marks **20a** and **20b** larger than the width W calculated by the above equation, it is possible to restrict the facet surface **60F** of the alignment mark **60a** from interfering with the alignment mark **60b**.

(43) Note that the actual width Wf of the facet surface **60F** may be larger than the calculated width W due to the influence of manufacturing conditions and the like. Therefore, it is preferable to set the interval P with a margin with respect to the calculated width W . However, if the interval P is increased more than necessary with respect to the calculated width W , the area occupied by the plurality of alignment marks **20** becomes large, which causes a decrease in the number of semiconductor devices manufactured from the SiC substrate **12**. In addition, when the area occupied by the plurality of alignment marks **20** becomes large, a photographing magnification when photographing the plurality of alignment marks with an alignment camera becomes small, and an alignment accuracy deteriorates. According to various experiments conducted by the inventors, it has been found that the actual width Wf of the facet surface **60F** does not reach ten times the calculated width W . Therefore, if the interval P is set so as to satisfy the relationship of $W<P<10W$ (that is, the relationship of $D/\tan \theta<P<10D/\tan \theta$), it is possible to restrict an interference of the facet surface **60F** of the alignment mark **60** with the adjacent alignment mark **60**

and an expansion of the area occupied by the alignment marks **20**. In the manufacturing method of the first embodiment, since the interval P satisfies the relationship of $W < P < 10W$, it is possible to restrict the interference of the facet surface **60F** of the alignment mark **60** with the adjacent alignment mark **60** and the expansion of the area occupied by the alignment marks **20**.

(44) In the manufacturing method of the first embodiment, an alignment process using the alignment marks **60** is performed after the epitaxial growth process. For example, alignment is performed using the alignment marks **60**, and etching, ion implantation, or the like may be performed on the SiC layer **50**, semiconductor layers other than the SiC layer **50**, or other layers such as electrode layers, insulating layers, or the like. For example, when ions are implanted into the grown SiC layer **50**, alignment may be performed using the alignment marks **60** to precisely form opening portions in an ion implantation mask, and ions may be implanted into the SiC layer **50** through the ion implantation mask. Since the facet surface **60F** of the alignment mark **60** does not interfere with other alignment marks **60**, alignment using the alignment marks **60** can be properly performed. Moreover, since the interval P between the alignment marks **60** is relatively narrow, the magnification can be increased when photographing the alignment marks **60** with the alignment camera. Therefore, the alignment can be performed with high accuracy.

(45) After that, the SiC substrate **12** is diced and divided into multiple pieces, thereby manufacturing multiple semiconductor devices.

(46) The semiconductor wafer (that is, SiC substrate **12**) illustrated in FIG. **16** has the following configuration. The semiconductor wafer (that is, the SiC substrate **12**) has the base substrate **12b** and an epitaxial layer (that is, the SiC layer **50**). The base substrate **12b** has the upper surface **12a** on which the alignment mark **20a** having the recessed shape is disposed. As illustrated in FIG. **2**, the perpendicular line **12s** that is perpendicular to the upper surface **12a** is inclined with respect to the direction toward the $[11-20]$ direction. As illustrated in FIG. **16**, the epitaxial layer (that is, the SiC layer **50**) is disposed on the upper surface **12a** of the base substrate **12b** and covers the alignment marks **20a**. On the upper surface **12a** of the base substrate **12b**, the alignment mark **20b** having the recessed shape is disposed as a structure. The structure (that is, the alignment mark **20b**) is arranged at a position apart from the alignment mark **20a** by the interval P in the $[11-20]$ direction along the upper surface **12a** of the base substrate **12b**. The interval P satisfies the relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is the depth of the alignment mark **20a** and θ is the inclination angle (that is, the off angle) of the perpendicular line **12s** that is perpendicular to the upper surface **12a** with respect to the $[0001]$ direction. Therefore, in this semiconductor wafer, the facet surface **60F** of the alignment mark **60a** does not interfere with the alignment mark **60b**, and proper alignment can be performed using the alignment mark **60a**. That is, by using this semiconductor wafer, semiconductor devices can be appropriately manufactured.

Second Embodiment

(47) In a manufacturing method of a second embodiment, semiconductor devices are manufactured from the SiC substrate **12** illustrated in FIGS. **1** and **2**, as in the first embodiment. In this manufacturing method, an alignment mark **20** having a recessed shape is formed on the upper surface **12a** of the SiC substrate **12** in a manner similar to that in the first embodiment. For example, as illustrated in FIG. **17**, an alignment mark **20** having a depth D smaller than a width W_a is formed. Next, as illustrated in FIG. **18**, a SiC layer **50** is epitaxially grown on the SiC substrate **12** (that is, the base substrate **12b**) so as to cover the alignment mark **20**. Accordingly, an alignment mark **60** following the shape of the alignment mark **20** is formed on an upper surface of the SiC layer **50**. In addition, a facet surface **60F** is formed at a position adjacent to the alignment mark **60** in the $[11-20]$ direction.

(48) Next, as illustrated in FIG. **19**, a structure **80** having a protruding shape is formed on the upper surface of the SiC substrate **12** (that is, the upper surface of the SiC layer **50**). Note that the structure **80** may be a conductor, an insulator, or a semiconductor. The structure **80** may be a part of a semiconductor device or a part of a mask. In the present embodiment, the structure **80** having a

protruding shape is formed at a position apart from the alignment mark **20** in the [11-20] direction. Between the alignment mark **20** and the structure **80**, an interval P is provided. The interval P is set so as to satisfy a relationship of $W < P < 10W$, that is, a relationship of $D/\tan \theta < P < 10D/\tan \theta$. Accordingly, interference between the structure **80** and the facet surface **60F** can be restricted without increasing the interval between the structure **80** and the alignment mark **20** more than necessary.

(49) After that, an alignment process using the alignment mark **60** is performed. Since the facet surface **60F** of the alignment mark **60** does not interfere with the structure **80**, alignment using the alignment mark **60** can be properly performed.

(50) After that, the SiC substrate **12** is diced and divided into multiple pieces, thereby manufacturing multiple semiconductor devices.

(51) The semiconductor wafer (that is, SiC substrate **12**) illustrated in FIG. **19** has the following configuration. The semiconductor wafer (that is, the SiC substrate **12**) has the base substrate **12b** and an epitaxial layer (that is, the SiC layer **50**). The base substrate **12b** has the upper surface **12a** on which the alignment mark **20** having the recessed shape is provided. As illustrated in FIG. **2**, the perpendicular line **12s** that is perpendicular to the upper surface **12a** is inclined with respect to the [0001] direction toward the [11-20] direction. As illustrated in FIG. **19**, the epitaxial layer (that is, the SiC layer **50**) is disposed on the upper surface **12a** of the base substrate **12b** and covers the alignment mark **20**. The structure **80** is disposed on the upper surface of the epitaxial layer (that is, the SiC layer **50**). The structure **80** is disposed at a position apart from the alignment mark **20** by the interval P in the [11-20] direction on the upper surface **12a** of the base substrate **12b**. The interval P satisfies the relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is the depth of the alignment mark **20a** and θ is the inclination angle (that is, the off angle) of the perpendicular line **12s** that is perpendicular to the upper surface **12a** with respect to the [0001] direction. Therefore, in this semiconductor wafer, the facet surface **60F** does not interfere with the structure **80**, and proper alignment can be performed using the alignment mark **60**.

(52) In the second embodiment described above, the structure **80** is a protruding portion disposed on the upper surface of the SiC layer **50**. However, as illustrated in FIG. **20**, the structure **80** may also be a recessed portion disposed on the upper surface of the SiC layer **50**. This configuration can also restrict interference between the structure **80** and the facet surface **60F**.

(53) In the first and second embodiments described above, each of the alignment marks **20** has a rectangular shape on the upper surface **12a** of the SiC substrate **12**. Each of the alignment mark **20** may also have other shapes. In these cases, a side surface of each of the alignment mark **20** located in the [11-20] direction does not have to be perpendicular to the [11-20] direction. For example, as illustrated in FIG. **21**, a side surface **24** (that is, an edge portion) of each of the alignment mark located in the [11-20] direction may extend along a direction diagonally intersecting the [11-20] direction.

(54) In the first and second embodiments described above, the SiC layer **50** is grown so that the thickness T of the SiC layer **50** is larger than the depth D of the alignment mark **20**. Alternatively, the SiC layer **50** may be grown so that the thickness T of the SiC layer **50** is less than the depth D of the alignment mark **20**. Even in this case, by setting the interval P so as to satisfy the relationship of $D/\tan \theta < P < 10D/\tan \theta$, interference of the facet surface **60F** with other structures can be restricted without increasing the interval P more than necessary.

(55) In the first and second embodiments described above, the alignment marks **20** are formed. Alternatively, the SiC substrate **12** formed in advance with the alignment marks **20** that satisfies the relationship of $D/\tan \theta < P < 10D/\tan \theta$ may be purchased and brought into a manufacturing facility. Even if the same epitaxial growth process and alignment process as in the first and second embodiments are performed on the SiC substrate **12** on which the alignment marks **20** are formed in advance, the same effect as in the first and second embodiments can be obtained.

(56) The SiC layer **50** in the first and second embodiments is an example of the epitaxial layer. The

alignment marks **20a**, **20** in the first and second embodiments are examples of the first alignment mark. The alignment mark **20b** in the first embodiment is an example of the structure having a recessed shape formed on the upper surface of the SiC substrate before forming the epitaxial layer. The structure **80** of the second embodiment is an example of the structure having a protruding shape or a recessed shape formed on the upper surface of the epitaxial layer. The alignment marks **60a** and **60** of the first and second embodiments are examples of the second alignment mark. (57) Although the embodiments have been described in detail above, these are merely examples and do not limit the scope of claims. The techniques described in the claims include various modifications and modifications of the specific examples illustrated above. The technical elements described in the present specification or the drawings exhibit technical usefulness alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. In addition, the techniques illustrated in the present specification or drawings achieve a plurality of objectives at the same time, and achieving one of the objectives itself has technical usefulness.

Claims

1. A manufacturing method of a semiconductor device, comprising: preparing a silicon carbide substrate that has an upper surface on which an alignment mark having a recessed shape is disposed, a perpendicular line that is perpendicular to the upper surface of the silicon carbide substrate being inclined with respect to a [0001] direction toward a [11-20] direction; growing an epitaxial layer on the upper surface of the silicon carbide substrate so as to cover the alignment mark; and forming a structure on or above the upper surface of the silicon carbide substrate, wherein the structure is formed at a position apart from the alignment mark by an interval P in the [11-20] direction along the upper surface of the silicon carbide substrate, and the interval P satisfies a relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is a depth of the alignment mark and θ is an inclination angle of the perpendicular line with respect to the [0001] direction.
2. The manufacturing method according to claim 1, wherein the forming the structure includes forming the structure having a recessed shape on the upper surface of the silicon carbide substrate before the growing the epitaxial layer.
3. The manufacturing method according to claim 1, wherein the forming the structure includes forming the structure having a protruding shape or a recessed shape on an upper surface of the epitaxial layer.
4. The manufacturing method according to claim 1, wherein when the upper surface of the silicon carbide substrate is viewed from above, an edge portion of the alignment mark located in the [11-20] direction is perpendicular to the [11-20] direction.
5. The manufacturing method according to claim 1, wherein the alignment mark is a first alignment mark, the growing the epitaxial layer includes forming a second alignment mark on an upper surface of the epitaxial layer, the second alignment mark having a recessed shape following the recessed shape of the first alignment mark, and the manufacturing method further comprising performing alignment using the second alignment mark.
6. A semiconductor wafer comprising: a base substrate made of silicon carbide and having an upper surface on which an alignment mark having a recessed shape is disposed, a perpendicular line that is perpendicular to the upper surface of the base substrate being inclined with respect to a [0001] direction toward a [11-20] direction; an epitaxial layer disposed on the upper surface of the base substrate and covering the alignment mark; and a structure disposed on the upper surface of the base substrate or an upper surface of the epitaxial layer, wherein the structure is disposed at a position apart from the alignment mark by an interval P in the [11-20] direction along the upper surface of the base substrate, and the interval P satisfies a relationship of $D/\tan \theta < P < 10D/\tan \theta$, where D is a depth of the alignment mark and θ is an inclination angle of the perpendicular line

with respect to the [0001] direction.

7. The semiconductor wafer according to claim 6, wherein the structure is a recessed portion disposed on the upper surface of the base substrate.

8. The semiconductor wafer according to claim 6, wherein the structure is a protruding portion or a recessed portion disposed on the upper surface of the epitaxial layer.

9. The semiconductor wafer according to claim 6, wherein when the upper surface of the base substrate is viewed from above, an edge portion of the alignment mark located in the [11-20] direction is perpendicular to the [11-20] direction.

10. The semiconductor wafer according to claim 6, wherein the alignment mark is a first alignment mark, and the upper surface of the epitaxial layer has a second alignment mark that has a recessed shape following the recessed shape of the first alignment mark.
